

Adaptive Sinhala Sign Language Learning System

25-26J-223

Final Report (Draft)

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B.Sc. (Hons) Degree in Information Technology Specialized in

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Department of Information Technology

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Declaration

I declare that this is my own work and this dissertation¹ does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Adaptive Sinhala Sign Language Learning System is an intelligent, mobile-based learning system designed for enhancing the efficiency and convenience of SSL learning. Conventional systems used for sign language training are not only based on static information but lack immediate feedback mechanisms and the ability to adapt according to the individual capabilities of learners, hence making them inefficient in providing quality training. In order to overcome these shortcomings, the proposed system utilizes artificial intelligence along with computer vision technology. The methodology involves real-time hand gesture recognition using camera input, where user-performed signs are analyzed against trained models to evaluate accuracy and generate confidence scores. The system incorporates adaptive learning techniques, including reinforcement learning, spaced repetition, and statistical feedback optimization, to tailor the learning path according to each user's performance and learning style. Multiple AI-driven agents collaborate to deliver customized feedback, recommend practice activities, and dynamically adjust difficulty levels. Testing was conducted by evaluating gesture recognition accuracy, response time, and user interaction efficiency. Results indicate that the system provides accurate real-time feedback and significantly enhances user engagement and learning retention. Users benefit from continuous progress tracking, targeted practice recommendations, and motivational feedback mechanisms. In summary, the above proposal has managed to change the process of learning sign language in that the learner is now able to learn on his/her own under guided intelligence. There could be other future developments involving dynamic signing and increased vocabulary range.

***Keywords:* Adaptive Learning, Sinhala Sign Language, Computer Vision, Reinforcement Learning, Personalized Education.**

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LIST OF ABBREVIATIONS

Sinhala Sign Language	SSL
Reinforcement Learning	RL
Artificial Intelligence	AI
Convolutional Neural Network	CNN
Application Programming Interface	API
User Interface	UI

1. INTRODUCTION

1.1 Background Literature

The rapid advancement of digital technologies has fundamentally reshaped modern education, giving rise to intelligent and interactive learning systems that significantly improve both accessibility and learning effectiveness. With the integration of artificial intelligence, mobile computing, and data-driven personalization, educational platforms are evolving from static content delivery systems into dynamic environments that actively respond to learner needs. Among these developments, sign language learning has emerged as an important research and application domain, particularly due to its role in enabling communication for deaf and hearing-impaired communities. Sinhala Sign Language (SSL), which is primarily used in Sri Lanka, is a visual-spatial language that relies on a combination of hand gestures, facial expressions, and body movements to convey meaning. Despite its critical importance for inclusive communication, SSL education remains relatively underdeveloped compared to globally recognized sign languages. This lack of technological support highlights the urgent need for innovative, intelligent, and localized solutions [7].

Traditional approaches to sign language learning are largely dependent on static educational resources such as printed textbooks, gesture charts, and pre-recorded instructional videos. While these resources provide foundational knowledge and serve as introductory tools, they lack interactivity and fail to engage learners in an active learning process. More importantly, they do not provide real-time feedback, which is essential for correcting gesture errors during practice. As a result, learners often struggle to determine whether their gestures are accurate, leading to repeated mistakes and slower learning progression. Furthermore, traditional learning methods typically follow a fixed curriculum structure that does not consider individual differences in learning pace, prior knowledge, or cognitive abilities. This one-size-fits-all approach reduces learner motivation and limits the overall effectiveness of the learning process. The absence of personalized feedback mechanisms and systematic progress tracking further exacerbates these limitations, making it difficult for learners to identify their strengths and weaknesses. In response to these challenges, computer vision has emerged as a powerful enabling technology for developing interactive sign language learning systems. Computer vision allows machines to interpret and analyze visual input captured through cameras, making it possible to detect and recognize hand gestures in real time. Techniques such as image processing, background segmentation, and hand landmark detection are widely used to extract meaningful features from gesture data. These features include finger positions, hand

orientation, and movement patterns, which are essential for accurate gesture recognition. By leveraging these techniques, modern systems can provide immediate visual feedback to users as they perform gestures, transforming the learning process from passive observation into active participation. This interactive capability significantly enhances user engagement and improves learning outcomes by enabling continuous practice and instant correction [9].

Machine learning, particularly deep learning, plays a crucial role in improving the accuracy and reliability of gesture recognition systems. Convolutional Neural Networks (CNNs) are extensively used for classifying static hand gestures due to their ability to automatically learn spatial hierarchies of features from images[8]. These models are trained using large, labeled datasets and can achieve high levels of accuracy in recognizing distinct hand shapes associated with different signs. For dynamic gesture recognition, where temporal information is critical, models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are employed. These models are capable of capturing sequential dependencies and understanding motion patterns over time, enabling the recognition of complex gestures that involve transitions and movement. The combination of CNNs for spatial feature extraction and sequence-based models for temporal analysis has significantly expanded the capabilities of sign language recognition systems, making them more robust and applicable to real-world scenarios. Another important advancement in educational technology is the concept of adaptive learning, which focuses on personalizing the learning experience for each user. Adaptive learning systems continuously monitor learner performance and adjust instructional content, difficulty levels, and feedback strategies accordingly. This approach is grounded in learner-centered educational theories, which emphasize the importance of addressing individual differences in learning styles, abilities, and progress rates. By analyzing user data such as accuracy, response time, and interaction patterns, adaptive systems can identify areas where learners struggle and provide targeted support. Conversely, they can accelerate learning for users who demonstrate strong performance, thereby optimizing the overall learning process. This level of personalization not only improves learning efficiency but also enhances user motivation and engagement. Reinforcement learning further strengthens adaptive learning systems by enabling them to optimize decision-making through continuous interaction with users. In this paradigm, the system acts as an agent that observes the learner's behavior, takes actions (such as adjusting difficulty or providing feedback), and receives rewards based on the effectiveness of those actions. Over time, the system learns which strategies yield the best learning outcomes and prioritizes them. For example, if a specific type of feedback consistently leads to improved user performance, the system will increasingly favor that feedback strategy. Additionally,

statistical approaches such as Bayesian inference and Thompson Sampling are used to model uncertainty and select optimal actions in dynamic environments. These techniques allow the system to balance exploration (trying new strategies) and exploitation (using known effective strategies), ensuring continuous improvement in personalization[12].

Spaced repetition is another key concept integrated into modern learning systems to enhance long-term knowledge retention. This technique is based on the psychological principle that information is more effectively retained when it is reviewed at strategically increasing intervals. By scheduling practice sessions based on the learner's forgetting curve, spaced repetition ensures that users revisit previously learned content at the optimal time for reinforcement. In the context of sign language learning, this approach helps learners retain gestures over the long term, reducing the likelihood of forgetting and improving overall proficiency. The integration of spaced repetition with adaptive learning further enhances its effectiveness, as review schedules can be dynamically adjusted based on individual performance. User engagement plays a vital role in determining the success of any learning system, particularly in skill-based domains such as sign language. Modern educational applications incorporate various motivational design elements, including progress tracking, achievement levels, and reward-based mechanisms, to encourage consistent user participation. Visual feedback mechanisms, such as correctness indicators, confidence scores, and instructional hints, provide immediate and meaningful information that guides learners during practice. These features not only make the learning process more interactive but also help maintain user motivation by providing a sense of accomplishment and progress. Research has shown that higher levels of engagement are directly correlated with improved learning outcomes, highlighting the importance of incorporating user-centered design principles. The widespread adoption of mobile technology has further accelerated the development of accessible learning solutions. Mobile-based learning systems offer flexibility and convenience, allowing users to engage in learning activities anytime and anywhere. This is particularly beneficial for sign language learners who may not have access to formal educational institutions or training centers. Mobile platforms also enable the integration of device-specific features such as cameras and sensors, which are essential for real-time gesture recognition[6]. As a result, mobile learning applications can provide a highly interactive and portable learning experience, reaching a broader audience and promoting continuous learning.

Despite these advancements, existing sign language learning systems still exhibit several limitations. A significant proportion of research and development efforts has been focused on widely used sign languages such as American Sign Language (ASL), leaving regional languages like Sinhala Sign Language relatively unexplored. This lack of focus results in limited availability of datasets, models, and educational tools tailored to SSL.

Furthermore, many existing systems either provide gesture recognition without adaptive learning capabilities or offer adaptive features without real-time feedback. This fragmentation reduces the overall effectiveness of these systems, as both real-time feedback and personalization are essential components of an effective learning experience. The proposed Adaptive Sinhala Sign Language Learning System addresses these gaps by integrating multiple advanced technologies into a unified platform. The system combines computer vision and deep learning techniques for accurate real-time gesture recognition with reinforcement learning-based adaptive mechanisms for personalized learning. By analyzing user performance continuously, the system dynamically adjusts lesson difficulty, feedback strategies, and review schedules. The inclusion of spaced repetition ensures long-term retention, while detailed performance tracking enables continuous improvement. Additionally, the system incorporates a user-centered design approach, providing an intuitive interface and interactive features that enhance usability and engagement.

In conclusion, the integration of intelligent technologies such as computer vision, machine learning, reinforcement learning, and adaptive learning has the potential to transform sign language education. These technologies overcome the limitations of traditional methods by providing interactive, personalized, and efficient learning experiences. The proposed system builds upon these advancements to deliver a comprehensive solution specifically designed for Sinhala Sign Language learners. By improving accessibility, engagement, and learning effectiveness, this system not only contributes to technological innovation but also promotes social inclusion by empowering individuals with effective communication skills.

1.2 Research Gap

Sign language learning systems have evolved significantly with the advancement of e-learning technologies, artificial intelligence, and smart computing. However, despite these developments, several limitations still exist in current systems, particularly in terms of personalization, adaptability, and real-time interaction. Existing research indicates that most systems focus on isolated functionalities rather than providing a comprehensive learning experience. Early approaches to sign language education were primarily based on multimedia content delivery. For instance, systems such as the one described in [1] present learning materials through synchronized video and text to support bilingual understanding. While this improves visual comprehension, it lacks interactivity and does not allow users to actively practice gestures. More importantly, these systems do not provide real-time feedback, making it difficult for learners to verify the correctness of

their gestures. As a result, learners may repeatedly practice incorrect signs, leading to slow progress and reduced learning effectiveness.

Similarly, more recent systems have incorporated smart computing techniques to enhance accessibility. The system presented in [2], for example, integrates gesture recognition and natural language processing to enable real-time translation. Although this improves communication, its primary objective is translation rather than learning. These systems rely heavily on pre-recorded outputs and do not evaluate user-generated gestures, which limit their usefulness in structured learning environments. Additionally, they do not guide learners through a progressive curriculum or provide performance-based recommendations.

A major limitation identified across existing systems is the absence of true adaptive learning. While some platforms include basic analytics to track user activity, they fail to dynamically adjust content based on individual performance. This results in a uniform learning experience that does not account for differences in user ability, learning speed, or error patterns. The lack of personalization significantly reduces learning efficiency and user engagement [4].

- Fixed learning paths that do not change based on performance
- No dynamic difficulty adjustment
- Limited use of user performance data for decision-making
- Inefficient learning progression for both beginners and advanced users

The difference between traditional and adaptive learning approaches can be summarized as follows:

Table 1: Comparison of Existing and Proposed Learning System Features

Feature	Existing Systems	Proposed System
Learning Path	Fixed	Dynamic and personalized
Difficulty Adjustment	Not supported	Based on performance
Feedback	Generic	Context-aware and adaptive
User Engagement	Moderate	High due to personalization

Another significant gap is related to gesture recognition capabilities. Most existing systems focus primarily on static gesture recognition, where individual hand signs are detected from single images. While this is suitable for basic alphabets, it does not reflect real-world sign language usage, which often involves dynamic gestures with motion and transitions. Many systems treat static and dynamic gestures separately, leading to

incomplete solutions that fail to capture the full complexity of sign language communication.

- Static gestures are easier to recognize but limited in expression
- Dynamic gestures require temporal analysis and are often ignored
- Lack of integration of both gesture types in a single system
- Reduced real-world applicability of learning systems

In addition to recognition limitations, feedback mechanisms in current systems are relatively weak. Most platforms provide only binary feedback, such as correct or incorrect, without offering detailed explanations or improvement suggestions [1]. This is insufficient for skill-based learning, where learners require guidance to refine their performance.

- No explanation of errors
- No corrective guidance for hand positioning or movement
- Lack of personalized feedback based on user history
- Minimal support for continuous improvement

Another important research gap is the limited application of reinforcement learning in sign language education. Reinforcement learning has the ability to enhance adaptive systems by learning from user interactions and optimizing decisions over time. However, existing systems rely mainly on predefined rules and static feedback mechanisms, which do not evolve based on user behavior.

- No learning from past user interactions
- Static feedback strategies
- No optimization of learning paths over time
- Lack of intelligent decision-making mechanisms

Furthermore, most existing systems are designed for widely used sign languages such as American Sign Language (ASL) or Greek Sign Language, with very limited focus on regional languages like Sinhala Sign Language. This creates a significant accessibility gap, as users who rely on SSL have fewer technological resources available for learning.

- Limited datasets for Sinhala Sign Language
- Lack of localized learning applications
- Minimal research focused on SSL
- Reduced accessibility for local users

Another critical issue is the lack of fully integrated systems. Existing solutions often specialize in a single functionality, such as recognition, translation, or content delivery,

but rarely combine all these features into one platform. This fragmentation results in inefficient learning experiences, as users must rely on multiple tools to achieve their goals.

Table 2: Evaluation of Existing Sign Language Systems

System Type	Strength	Limitation
Recognition Systems	Accurate gesture detection	No learning support
Translation Systems	Real-time communication	No skill development
Learning Platforms	Content delivery	No real-time interaction

The proposed Adaptive Sinhala Sign Language Learning System addresses these limitations by integrating multiple advanced technologies into a single platform[10][11]. It introduces a personalized learning path that dynamically adjusts based on user performance, ensuring that learners receive appropriate guidance at every stage. The system combines both static and dynamic gesture recognition, enabling a more comprehensive understanding of sign language. Additionally, it incorporates reinforcement learning to optimize feedback strategies, allowing the system to improve continuously through user interaction.

- Personalized and adaptive learning experience
- Real-time gesture recognition and feedback
- Integration of static and dynamic gesture analysis
- Reinforcement learning-based decision-making
- Focus on Sinhala Sign Language for improved accessibility

By addressing these gaps, the proposed system transforms sign language learning from a passive activity into an interactive, intelligent, and user-centered process. It provides learners with real-time feedback, adaptive guidance, and continuous performance improvement, ultimately enhancing both learning efficiency and user engagement. In conclusion, while existing systems have contributed to the advancement of sign language technology, they fall short in delivering a fully adaptive and comprehensive learning environment. The proposed system bridges these gaps by combining recognition, personalization, and intelligent feedback into a unified solution, thereby contributing to both technological innovation and inclusive education.

1.3 Research Problem

Effective communication is a fundamental human need, yet individuals with hearing impairments continue to face significant barriers in accessing quality education and communication tools. Sign language serves as a primary medium of communication for the deaf community; however, learning sign language remains a complex and resource-intensive process. In particular, Sinhala Sign Language (SSL) lacks sufficient technological support compared to widely studied sign languages, resulting in limited accessibility to structured and effective learning systems. Traditional sign language learning techniques mostly use static content for learning like books, charts, and recorded video lessons. These learning methods offer only static content related to the subject, but they lack any interactive process through which the learners can practice the gestures. The learners cannot evaluate whether they are making correct gestures in sign languages or not; therefore, it is common that learners may develop bad habits in learning the technique. Sign language learning systems, like an e-learning system presented in [1], allow learners to learn in two languages through sign language videos; however, these learning tools fail to support active learning.

New developments in artificial intelligence and smart computing have led to new opportunities for sign language solutions. One such system described in [2] utilizes techniques from natural language processing and gesture detection, allowing for real-time sign language translation. While these solutions provide greater access and communication capabilities, they are focused on translating sign language as opposed to learning it. This means that user progress and skill development are not tracked in any way. Thus, there is no provision in place for improving user knowledge and proficiency in sign language. A critical issue in current sign language learning systems is the lack of personalized learning paths. Most systems provide uniform content to all users, regardless of their skill level, learning speed, or prior knowledge. This one-size-fits-all approach does not align with modern educational principles, which emphasize learner-centered design and adaptive instruction. Without personalization, learners may either struggle with content that is too advanced or lose interest in content that is too basic. This significantly reduces the effectiveness of the learning process and highlights the need for adaptive learning mechanisms [3], [4].

Another major research problem is the limited capability of existing systems to handle both static and dynamic sign recognition. Many systems focus solely on static gestures, such as alphabet signs, which can be detected from single images. However, real-world sign language communication involves dynamic gestures that include motion, timing, and

transitions between signs. Some advanced systems attempt to address dynamic recognition using sequence-based models, but these are often implemented separately from static recognition systems [5], [6]. The lack of an integrated approach results in incomplete learning solutions, preventing users from achieving comprehensive sign language proficiency.

Apart from the issues surrounding recognition, there are also problems regarding feedback mechanisms that have been incorporated into many of the current systems. Feedback has been limited to simple information on whether the gesture has been performed correctly or incorrectly. Such an approach fails to offer any guidance in improving one's performance or correcting any mistakes made. To learn effectively, it is necessary to have feedback mechanisms that allow one to learn more about mistakes and improve on their skills [7]. Reinforcement learning (RL) has shown great potential in adaptive systems by enabling continuous learning and optimization based on user interactions. RL-based approaches allow systems to dynamically adjust feedback and recommendations to maximize learning outcomes. However, the application of reinforcement learning in sign language education remains limited. Existing systems rarely utilize RL to personalize feedback or optimize learning strategies over time [8]. This gap presents an opportunity to enhance the effectiveness of sign language learning systems through intelligent, data-driven feedback mechanisms.

Furthermore, most research in sign language technology has focused on globally recognized languages such as American Sign Language (ASL), with limited attention given to regional languages like Sinhala Sign Language. This creates a significant disparity in accessibility and technological support. The lack of datasets, models, and research specific to SSL further complicates the development of effective learning systems. Addressing this issue is essential for promoting inclusivity and ensuring equal learning opportunities for all users [1], [9]. Another important problem is the lack of fully integrated systems that combine gesture recognition, adaptive learning, and intelligent feedback into a single platform. Existing solutions often address these components independently, resulting in fragmented systems that do not provide a seamless learning experience. Learners are required to use multiple tools, which can be inefficient and confusing. A unified system that integrates all these features is necessary to support comprehensive and effective learning [4], [7].

The proposed Adaptive Sinhala Sign Language Learning System is designed to address these research problems by introducing several key innovations. First, it implements a personalized learning path that adapts to individual user performance, ensuring that

learners receive appropriate content and guidance. Second, it integrates both static and dynamic gesture recognition within a single system, enabling a more complete understanding of sign language. Third, it incorporates reinforcement learning to provide intelligent, adaptive feedback that evolves based on user interactions. By addressing the limitations of existing systems, the proposed solution aims to create a more effective, engaging, and accessible learning environment. It transforms sign language learning into an interactive process where users actively participate, receive real-time feedback, and follow personalized learning paths. Additionally, by focusing on Sinhala Sign Language, the system contributes to reducing the research gap in regional sign language technologies and promotes inclusivity in education.

In conclusion, the key research problem lies in the lack of an integrated, adaptive, and intelligent sign language learning system that supports personalized learning, comprehensive gesture recognition, and advanced feedback mechanisms. Existing systems fall short in addressing these aspects, highlighting the need for a novel approach. The proposed system aims to overcome these challenges by leveraging modern technologies to enhance learning outcomes and accessibility for SSL learners.

1.4 Objectives

The primary aim of this research is to design and develop an Adaptive Sinhala Sign Language Learning System that enhances the effectiveness, accessibility, and interactivity of sign language education. This system focuses on integrating advanced technologies such as computer vision, machine learning, and reinforcement learning to provide a personalized and intelligent learning experience[1]. The objectives of this study are structured to address the limitations identified in existing systems and to introduce innovative solutions that improve learning outcomes for users. The main objective of this research is to develop a comprehensive and adaptive learning platform for Sinhala Sign Language (SSL) that supports both beginners and intermediate learners. Unlike traditional learning methods, which rely on static resources, this system aims to create an interactive environment where users can actively practice sign language and receive real-time feedback. By enabling users to engage directly with the system, the learning process becomes more efficient, practical, and user-centered[2].

A key objective of this research is to implement a personalized learning path that adapts to individual user performance. Each learner has unique abilities, learning speeds, and prior knowledge, which must be considered to ensure effective learning. The proposed system aims to analyze user interactions, track performance, and dynamically adjust learning content based on individual needs. This includes recommending appropriate lessons, adjusting difficulty levels, and focusing on areas where the user requires improvement. By incorporating adaptive learning principles, the system ensures that learners are neither overwhelmed nor under-challenged, leading to improved engagement and retention [1]. Another important objective is to develop a robust gesture recognition module capable of detecting both static and dynamic sign language gestures. Static gestures, such as alphabet signs, are relatively easier to detect using image-based models. However, dynamic gestures involve motion and temporal sequences, making them more complex to recognize. This research aims to integrate both static and dynamic recognition into a single unified system. By doing so, learners can practice a wider range of signs, from simple alphabets to more complex gestures, thereby improving their overall communication skills. The combination of these two recognition approaches ensures a more complete and realistic learning experience.

In addition to gesture recognition, providing effective feedback is a critical objective of this research. Traditional systems often provide basic feedback, such as indicating whether a gesture is correct or incorrect. However, this type of feedback does not guide users on how to improve. The proposed system aims to provide detailed and meaningful feedback, including confidence scores, error identification, and suggestions for improvement[12].

This helps learners understand their mistakes and refine their gestures more effectively, leading to faster skill development. To further enhance the feedback mechanism, this research aims to incorporate reinforcement learning (RL) into the system. Reinforcement learning allows the system to learn from user interactions and optimize feedback strategies over time. By analyzing how users respond to different types of feedback, the system can determine which approach is most effective for improving learning outcomes. This enables the system to provide more personalized and intelligent guidance, making the learning experience more efficient and adaptive [2].

Another objective of this research is to improve user engagement through interactive and user-friendly system design. Engagement plays a vital role in learning, especially in skill-based domains such as sign language. The system aims to include features such as progress tracking, performance visualization, and interactive feedback to motivate users. By providing a clear view of their progress and achievements, learners are encouraged to practice consistently and stay engaged with the system. Accessibility is also a major focus of this research. The system is designed to be accessible to a wide range of users, including those who may not have access to formal sign language education. By developing a mobile or web-based platform, the system ensures that users can learn anytime and anywhere. This flexibility makes the learning process more convenient and inclusive, particularly for individuals in remote or underserved areas.

Another important objective is to address the lack of technological support for Sinhala Sign Language. Most existing systems focus on widely used sign languages such as American Sign Language, leaving regional languages underrepresented. This research aims to contribute to the development of SSL by creating a system specifically tailored to its structure and requirements. This not only improves accessibility for local users but also contributes to the preservation and promotion of the language [3].

Furthermore, this research aims to develop an integrated system that combines gesture recognition, adaptive learning, and intelligent feedback into a single platform. Many existing solutions address these components separately, resulting in fragmented learning experiences. By integrating all these features, the proposed system provides a seamless and efficient learning environment where users can practice, receive feedback, and track their progress within a single application. Another objective is to evaluate the performance and effectiveness of the proposed system. This includes measuring gesture recognition accuracy, system response time, and user satisfaction. By conducting testing and analysis, the research aims to validate the effectiveness of the system in improving learning outcomes. User feedback will also be considered to identify areas for further improvement. In addition, the system aims to support continuous learning and long-term

skill development. Learning sign language requires consistent practice and reinforcement over time. The proposed system includes mechanisms such as performance tracking and adaptive recommendations to ensure that users continue to improve their skills. This long-term approach helps learners achieve mastery rather than just basic understanding.

Finally, this research aims to contribute to the field of intelligent educational systems by demonstrating how advanced technologies can be applied to sign language learning. By integrating computer vision, machine learning, and reinforcement learning, the study provides insights into the development of adaptive and interactive learning platforms. The findings of this research can be extended to other domains, contributing to the broader advancement of educational technology.

In conclusion, the objectives of this research are centered on developing an adaptive, interactive, and intelligent learning system for Sinhala Sign Language. The focus on personalized learning paths, combined static and dynamic gesture recognition, and reinforcement learning-based feedback distinguishes this system from existing solutions. By addressing current limitations and introducing innovative features, the proposed system aims to enhance learning effectiveness, improve accessibility, and promote inclusivity in sign language education.

2. METHODOLOGY

2.1 Overview

This study proposes an Adaptive Sinhala Sign Language Learning System designed to enhance the effectiveness of sign language education through intelligent personalization. Traditional sign language learning systems often rely on static lesson structures, generic feedback, and limited mechanisms for long-term knowledge retention. These limitations reduce user engagement and slow down the learning process. To address these challenges, the proposed system integrates machine learning for sign recognition, reinforcement learning for adaptive lesson delivery, and spaced repetition techniques for memory retention. The system is capable of analyzing user performance in real time and dynamically adjusting lesson difficulty, feedback, and review intervals. This enables a personalized learning experience tailored to individual user needs, improving both engagement and learning outcomes.

2.2 System Architecture and Design

The system follows a client-server architecture, consisting of three main layers:

- **Mobile Application Layer:** Provides the user interface and captures real-time video input using the device camera.
- **Backend Server Layer:** Handles data processing, machine learning inference, and adaptive decision-making using a Flask-based API.
- **Data Storage Layer:** Stores user performance history, model parameters, and adaptive learning data.

The mobile application communicates with the backend via RESTful APIs, enabling real-time interaction and feedback. The backend integrates multiple components, including preprocessing modules, sign recognition models, reinforcement learning agents, and performance tracking systems.

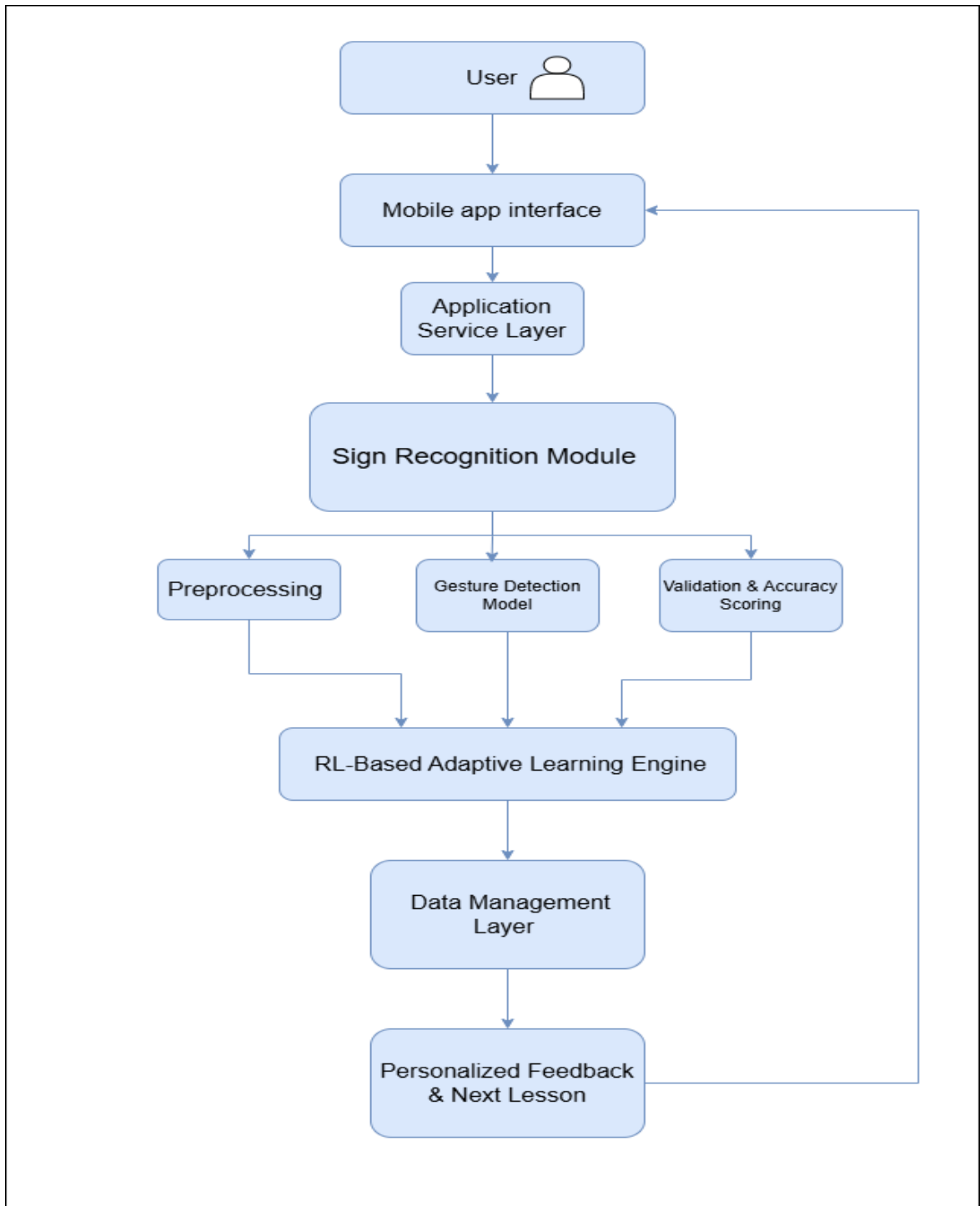


Figure 1: System Diagram

2.2.1 Component Description

The system is composed of several key functional components:

- **Input Data Pipeline:** The system captures real-time video or image input from the user. Basic validation checks are performed to ensure adequate lighting, hand visibility, and image quality before further processing.
- **Preprocessing Module:** Captured images are normalized to ensure consistency across varying environmental conditions. Techniques such as contrast enhancement and brightness normalization are applied, followed by resizing images to a standard format suitable for model input.
- **Sign Recognition Model:** A Convolutional Neural Network (CNN) is used to classify Sinhala sign language gestures. The model outputs the predicted sign along with a confidence score, which is later used in adaptive decision-making.
- **Adaptive Learning Engine:** A reinforcement learning-based module analyzes user performance and determines the next learning action. It adjusts difficulty levels, selects appropriate feedback, and schedules future practice sessions.
- **Spaced Repetition Module:** This module determines optimal review intervals based on user performance, ensuring long-term retention of learned signs.
- **Performance Tracking System:** The system records user interactions, including accuracy, response time, and learning progress, enabling continuous personalization.

2.2.2 Frontend System Design and Implementation

The frontend of the proposed system is implemented as a mobile-based interactive application designed to facilitate real-time Sinhala sign language learning. It is developed using React Native with Expo, enabling cross-platform compatibility and efficient performance on mobile devices. The frontend acts as the primary interface between the user and the system, handling user interaction, gesture capture, feedback visualization,

and learning progression. It is designed following modular architecture to ensure scalability, maintainability, and responsiveness.

Architecture and Design Approach

The frontend follows a layered design structure consisting of:

Table 3: Description of Frontend Architectural Layers and Responsibilities

Presentation Layer	Responsible for rendering the user interface and displaying learning content, feedback, and progress indicators
Logic Layer	Handles lesson flow, gesture verification triggers, and interaction control
State Management Layer	Manages user progress, session data, and learning states using Context API and local state
Data Persistence Layer	Stores user progress and performance data locally using AsyncStorage
Integration Layer	Communicates with the backend via RESTful APIs and accesses device camera for gesture input

This separation ensures clear responsibility distribution and improves system maintainability.

- **Component Design Strategy**

The frontend is developed using component-based architecture. Two main types of components are used:

- **Container Components:** Manage application logic, including gesture verification, API communication, and navigation.
- **Presentational Components:** Handle UI rendering and user interaction without containing business logic.

Reusable logic is encapsulated by using custom hooks, enabling efficient handling of:

- Letter progression
- Gesture detection and verification
- Feedback processing

This approach improves modularity and reusability across the application.

- **State Management and Data Handling**

The system uses a hybrid state management approach:

Table 4: State Management Approaches in the System

State Management Method	Purpose	Example Usage in the system
Local State	used for temporary UI states such as loading indicators and animations.	Button loading spinner, form input states, animation trigger
Context API	manages global learning data such as levels, progress, and user performance	User level tracking, lesson progress, performance scores across screens
AsyncStorage	used for persistent storage of user progress, attempt history, and performance metrics	Saving completed lessons, quiz history, and long-term learning data

This ensures that user data is retained across sessions and supports adaptive learning.

- **Real-Time Feedback Integration**

The frontend enables real-time interaction by capturing user gestures through the device camera and sending them to the backend for processing. Based on the prediction results:

- Correct gestures trigger positive visual feedback and progression to the next task.
- Incorrect gestures trigger adaptive feedback generated by the reinforcement learning module.

The feedback is displayed using interactive UI elements, ensuring immediate and clear communication with the user.

- **Performance Optimization**

To ensure smooth operation on mobile devices, several optimization techniques are applied:

- Frame skipping during real-time camera processing
- Image compression before transmission to the backend
- Efficient rendering using optimized list components
- Use of native animation drivers for smooth transitions

These optimizations reduce computational load and improve responsiveness.

2.3 Data Collection and Preprocessing

2.3.1 Dataset Collection

The effectiveness of the sign recognition model depends heavily on the quality and diversity of the dataset used for training. In this study, a dataset of Sinhala sign language gestures was created using a structured data collection process. The dataset consists of images representing 25 Sinhala letters, covering both vowels and consonants. Each sign was captured multiple times under control conditions to ensure consistency. The data collection process involved the use of a camera-based system to record hand gestures from different individuals. This approach allowed the dataset to capture variations in hand size, orientation, and positioning. To maintain consistency, images were captured against a plain background, minimizing distractions and improving the clarity of hand gestures. Lighting conditions were carefully controlled to reduce shadows and reflections. Additionally, multiple samples were collected for each sign to ensure that the model could generalize effectively across different scenarios. The dataset was organized into separate categories, with each category corresponding to a specific sign. This structured organization facilitated efficient training and evaluation of the machine learning model.

2.3.2 Preprocessing Techniques

Preprocessing plays a critical role in preparing raw input data for machine learning. The primary objective of preprocessing is to enhance image quality, reduce variability, and ensure compatibility with the model. One of the key techniques used in this study is

contrast enhancement, which improves the visibility of hand features in different lighting conditions. This step ensures that important details are not lost due to poor illumination.

Another important technique is brightness normalization, which adjusts the intensity of images to a consistent level. This helps reduce the impact of varying lighting conditions on model performance. All images are then resized to a standard resolution, ensuring uniformity across the dataset. This step is essential for maintaining consistency during model training and inference.

In addition to these techniques, the system also extracts hand landmarks and positional features using advanced computer vision methods. These features provide additional information about hand structure and movement, improving the accuracy of sign recognition.

For video-based inputs, multiple frames are analyzed to reduce prediction instability. By considering a sequence of frames rather than a single image, the system can produce more reliable predictions.

2.3.3 Data Augmentation

To improve the robustness of the model, data augmentation techniques are applied during training. These techniques artificially increase the diversity of the dataset by introducing variations such as rotation, brightness adjustment, and slight distortions. Data augmentation helps the model become more resilient to real-world conditions, such as changes in lighting, hand orientation, and background noise. As a result, the system performs more effectively when deployed in practical environments.

2.4 Machine Learning Model for Sign Recognition

The accurate recognition of hand gestures is a fundamental requirement for the effectiveness of the proposed Adaptive Sinhala Sign Language Learning System. In this study, a deep learning-based approach is utilized to identify and classify Sinhala sign language gestures in real time. The system employs a Convolutional Neural Network (CNN), which is widely recognized for its superior performance in image classification tasks. The CNN model is designed to learn spatial features from input images, enabling it to distinguish between different hand shapes and movements associated with Sinhala sign language. The input to the model consists of preprocessed images, which are normalized and resized to a consistent format. These images are then passed through multiple

convolutional layers, where feature extraction takes place. Each layer captures increasingly complex patterns, such as edges, shapes, and hand configurations.

Following the convolutional layers, pooling operations are applied to reduce the spatial dimensions of the feature maps. This not only decreases computational complexity but also helps in retaining the most significant features. The extracted features are then flattened and passed through fully connected layers, which perform the final classification. The output layer of the model uses a SoftMax activation function to generate a probability distribution over all possible sign classes. The class with the highest probability is selected as the predicted sign. In addition to the predicted label, the model provides a confidence score, which indicates the certainty of the prediction. This confidence plays a critical role in subsequent adaptive decision-making processes. To enhance the robustness of the system, the model is trained on a diverse dataset that includes variations in hand orientation, lighting conditions, and user differences. Furthermore, techniques such as data augmentation are used during training to improve generalization and reduce overfitting.

In addition to static image recognition, the system is capable of handling dynamic gestures by analyzing sequences of frames. Temporal consistency is achieved by evaluating multiple consecutive frames and selecting the most frequent prediction. This approach reduces noise and improves the stability of predictions.

Overall, the machine learning model serves as the foundation of the system, enabling accurate and reliable recognition of Sinhala sign language gestures, which is essential for providing meaningful feedback and adaptive learning

2.5 Reinforcement Learning-Based Adaptive Learning

One of the key innovations of this research is the integration of reinforcement learning (RL) to create a personalized and adaptive learning experience. Unlike traditional systems that follow a fixed curriculum, the proposed system continuously learns from user interactions and dynamically adjusts its behavior to optimize learning outcomes. Reinforcement learning is a type of machine learning in which an agent learns to make decisions by interacting with an environment. The agent observes the current state, takes an action, and receives a reward based on the outcome. Over time, the agent learns to select actions that maximize cumulative rewards.

In the context of this system, the learner is modeled as the environment, and the adaptive engine acts as the agent. The agent observes the learner's performance and determines the most appropriate next step in the learning process.

2.5.1 State Representation

To effectively personalize the learning experience, the system must maintain a comprehensive understanding of the learner's current status. This is achieved through a multi-dimensional state representation, which captures various aspects of user performance.

The state is defined using the following parameters:

- **Accuracy:** Represents the proportion of correctly performed signs over recent attempts. It reflects the learner's level of understanding.
- **Confidence Score:** Indicates the certainty of the model's predictions. High confidence suggests that the sign was performed correctly and clearly.
- **Response Time:** Measures how quickly the learner performs a sign. Faster responses generally indicate better familiarity.
- **Streak:** Tracks consecutive correct attempts, providing insight into consistency.
- **Recency:** Represents the time elapsed since the last practice session for a particular sign.
- **Difficulty Level:** Indicates the current level of complexity assigned to the learner.

These parameters are discretized into categories to create a manageable state space. By combining these factors, the system obtains a holistic view of the learner's progress and challenges.

2.5.2 Action Space

Based on the current state, the reinforcement learning agent selects an action that determines how the system should respond. The action space is designed to influence multiple aspects of the learning process simultaneously.

The main actions include:

- **Difficulty Adjustment:** The system can increase, decrease, or maintain the current difficulty level. This ensures that the learner is always working at an appropriate level of challenge.

- **Feedback Selection:** Different types of feedback can be provided, including positive reinforcement, constructive guidance, or motivational messages.
- **Review Scheduling:** The system determines when the learner should revisit a particular sign, supporting long-term retention.

By combining these actions, the system creates a fully adaptive learning pathway tailored to each individual user.

2.5.3 Reward Mechanism

The effectiveness of the reinforcement learning model depends on the design of the reward function. In this study, a multi-objective reward mechanism is used to evaluate the quality of actions taken by the agent.

The reward is composed of several components:

Accuracy-Based Reward

Correctly performed signs are rewarded positively, while incorrect attempts receive penalties. This encourages the system to guide users toward correct learning.

Improvement-Based Reward

The system evaluates whether the learner's performance has improved compared to previous attempts. Positive improvement results in additional rewards, promoting continuous progress.

Retention-Based Reward

The ability to recall previously learned signs after a time interval is rewarded. This ensures that the system prioritizes long-term learning rather than short-term success.

Engagement-Based Reward

User engagement is also considered. Consistent practice and longer interaction sessions are rewarded, while early session termination may result in penalties.

By combining these components, the reward function ensures a balanced approach that promotes accuracy, improvement, retention, and engagement.

2.5.4 Learning Process

The reinforcement learning agent updates its knowledge using a Q-learning algorithm, which estimates the expected reward for each state-action pair. Over time, the agent learns which actions produce the best outcomes for different learner states.

Initially, the agent explores different actions to gather information about their effects. As more data is collected, the agent gradually shifts toward exploiting the best-known strategies. This balance between exploration and exploitation ensures that the system continuously improves its decision-making.

The learned knowledge is stored in a structured format, allowing the system to quickly retrieve optimal actions for any given state. As more users interact with the system, the model becomes increasingly effective at personalization.

2.5.5 Personalized Learning Path

One of the most significant advantages of using reinforcement learning is the ability to generate personalized learning paths. Unlike fixed curricula, the system dynamically selects the next sign or activity based on the learner's current state.

For example:

- A learner with high accuracy may be introduced to more complex signs.
- A learner struggling with a particular sign may receive additional practice and simplified tasks.
- Frequently forgotten signs are prioritized for review.

This adaptive approach ensures that each learner progresses at their own pace, maximizing efficiency and minimizing frustration.

2.5.6 Handling Static and Dynamic Signs

The system is designed to support both static and dynamic sign recognition, which adds complexity to the learning process. Static signs involve fixed hand shapes, while dynamic signs include motion and transitions.

The reinforcement learning model accounts for this distinction by adjusting difficulty levels and feedback strategies accordingly. For dynamic signs, additional emphasis is

placed on temporal accuracy and motion consistency. This ensures that learners receive appropriate guidance for different types of gestures.

2.6 Curriculum and Learning Strategy

An effective learning system requires a well-structured curriculum that guides learners from basic concepts to more advanced levels. In traditional sign language learning platforms, the curriculum is typically fixed, offering the same sequence of lessons to all users. However, such an approach does not account for individual differences in learning pace, prior knowledge, and skill level. To address this limitation, the proposed system incorporates both static curriculum design and dynamic adaptation, enabling a balance between structured learning and personalization.

2.6.1 Static Curriculum Structure

The system organizes Sinhala sign language learning into multiple levels, each representing a specific stage of difficulty. The progression begins with simple signs and gradually introduces more complex gestures. At the initial stage, learners are introduced to basic vowel signs, which involve simple hand shapes and minimal movement. These signs help users become familiar with hand positioning and gesture formation. As learners progress, they are introduced to basic consonants, which require slightly more coordination and variation in hand movements.

Intermediate levels include combinations of hand shapes and orientations, requiring greater precision and control. Advanced levels introduce complex and dynamic signs, which involve motion, transitions, and more intricate hand configurations. Each level is associated with a target performance threshold, such as a minimum accuracy level that must be achieved before progressing to the next stage. This ensures that learners develop a solid understanding before moving forward.

2.6.2 Dynamic Difficulty Adaptation

While the static curriculum provides structure, the system's true strength lies in its ability to dynamically adapt the learning experience based on user performance.

The reinforcement learning agent continuously monitors the learner's progress and adjusts the difficulty level accordingly. If a learner demonstrates consistent accuracy and confidence, the system increases the difficulty by introducing more challenging signs or

reducing support. On the other hand, if the learner struggles, the system reduces the difficulty by providing simpler tasks, additional practice, and more detailed guidance.

This adaptive mechanism prevents learners from experiencing frustration due to overly difficult tasks or boredom due to repetitive and easy exercises. Instead, it maintains an optimal level of challenge, which is essential for effective learning.

2.6.3 Learning Path Optimization

The system dynamically selects the next learning activity based on multiple factors, including performance history, current difficulty level, and review requirements. This results in a personalized learning path that differs for each user. For example, if a learner has mastered a particular sign, the system may introduce a new sign or increase the difficulty level. Conversely, if a learner repeatedly struggles with a specific sign, the system may prioritize that sign for additional practice. This approach ensures that learning is both efficient and targeted, focusing on areas that require improvement while reinforcing previously learned content.

2.7 Feedback and Personalization

Providing effective feedback is essential for guiding learners and improving performance. In many traditional systems, feedback is limited to simple correctness indicators, which do not provide sufficient guidance for improvement. The proposed system addresses this limitation by implementing a personalized feedback mechanism.

2.7.1 Types of Feedback

The system generates different types of feedback based on user performance:

- **Positive Feedback**
When a learner performs a sign correctly, the system provides encouraging messages to reinforce correct behavior and boost confidence.
- **Constructive Feedback**
For incorrect attempts, the system provides specific guidance on how to improve. This may include suggestions related to hand positioning, movement, or timing.
- **Motivational Feedback**
To maintain user engagement, the system provides motivational messages, especially when learners face difficulties or show signs of reduced activity.

2.7.2 Context-Aware Feedback Selection

The reinforcement learning agent plays a key role in selecting the most appropriate feedback. Instead of using a fixed set of responses, the system analyzes the learner's state and chooses feedback that is most likely to be effective.

For example:

- Beginners may receive more detailed guidance and encouragement
- Intermediate learners may receive balanced feedback
- Advanced learners may receive minimal feedback focused on refinement

This context-aware approach ensures that feedback is relevant, meaningful, and supportive.

2.7.3 Enhancing User Engagement

User engagement is a critical factor in the success of any learning system. The proposed system incorporates several strategies to maintain high levels of engagement. The use of visual feedback, progress tracking, and performance indicators helps learners understand their progress and stay motivated. Additionally, the adaptive nature of the system ensures that learners are continuously challenged without being overwhelmed. By providing timely and personalized feedback, the system encourages consistent practice, which is essential for skill development.

2.7.4 Continuous Personalization

Personalization is not a one-time process but an ongoing cycle. The system continuously collects data from user interactions and updates its models accordingly. This allows the system to refine its understanding of each learner and improve its recommendations over time. As a result, the learning experience becomes increasingly tailored to individual needs, leading to better outcomes and higher satisfaction.

2.8 Commercialization Aspect of the Product

2.8.1 Product Overview

This product design aims at developing an adaptive Sinhala sign language learning system as a mobile-based platform that can promote learning in a way that accommodates those

who have taken interest in learning Sinhala sign language. It aims at meeting the requirements of learners, instructors, people with different abilities, and organizations who need such a tool for communication purposes.

2.8.2 Target Market

The primary target users include:

- Students learning Sinhala sign language
- Individuals with hearing or speech impairments
- Educational institutions and special education centers
- NGOs and organizations supporting inclusive communication

The secondary market includes general users interested in learning sign language and improving communication skills.

2.8.3 Value Proposition

The system provides several key advantages over existing solutions:

- Personalized learning through reinforcement learning
- Real-time gesture recognition using AI
- Adaptive feedback and difficulty adjusting
- Offline-supported learning features
- User-friendly mobile interface

These features ensure improved learning efficiency, better engagement, and long-term knowledge retention.

2.8.4 Revenue Model

The commercialization strategy can adopt a freemium model, which includes:

- Free Version: Basic lessons and limited features

- Premium Version: Advanced lessons, personalized analytics, and full access to adaptive learning features

Additional revenue streams may include:

- Institutional licensing for schools and training centers
- Partnerships with educational organizations
- Sponsored content or government-funded accessibility programs

2.8.5 Deployment Strategy

The system can be deployed as a mobile application using platforms such as:

- Android (Google Play Store)
- iOS (Apple App Store)

Cloud-based backend services ensure scalability and real-time processing. Regular updates and feature enhancements will be delivered through application updates.

2.8.6 Future Expansion

Future commercialization opportunities include:

- Support for additional sign languages
- Integration with wearable devices
- Expansion into global accessibility platforms
- Collaboration with government and educational sectors

2.9 Testing and Implementation

2.9.1 Implementation Overview

The system is implemented using a client-server architecture:

- **Frontend:** Developed using React Native and Expo for cross-platform mobile compatibility

- **Backend:** Implemented using Flask to handle API requests, machine learning inference, and adaptive logic
- **Database:** Stores user progress, performance data, and learning history

The frontend handles user interaction, camera input, and feedback display, while the backend processes gesture recognition and adaptive learning decisions.

2.9.2 System Integration

All system components are integrated through RESTful APIs:

- The mobile application captures user gestures via the camera
- Data is sent to the backend for processing
- The machine learning model predicts the gesture
- The reinforcement learning module determines feedback and next actions
- Results are returned to the frontend for real-time feedback display

This ensures smooth communication between all modules and real-time system responsiveness.

2.9.3 Testing Strategy

- **Unit Testing**

Individual components such as preprocessing functions, API endpoints, and frontend modules are tested independently to ensure correctness.

- **Integration Testing**

Integration testing verifies the interaction between:

- Frontend and backend APIs
- Machine learning model and preprocessing module
- Reinforcement of learning module and feedback system

- **System Testing**

The complete system is tested as a whole to ensure all functionalities work together correctly, including:

- Gesture detection accuracy
- Feedback generation
- Adaptive learning behavior

- **User Testing**

User testing is conducted with real users to evaluate:

- Usability and interface design
- Learning effectiveness
- User satisfaction and engagement

Feedback from users is used to improve system performance and usability.

- **Performance Evaluation**

The system is evaluated based on:

- Accuracy: Correct classification of sign gestures
- Response Time: Speed of real-time feedback
- User Engagement: Frequency and duration of usage
- Learning Improvement: Progress in user performance over time

These metrics help assess the effectiveness of the system in real-world scenarios.

2.10 Implementation Challenges

Some key challenges encountered include:

- Variations in lighting and background conditions
- Differences in hand shapes and gesture styles
- Real-time processing constraints on mobile devices

These challenges are addressed through preprocessing techniques, data augmentation, and optimization strategies.

3. RESULTS & DISCUSSION

3.1 Results

The proposed Adaptive Sinhala Sign Language Learning System was implemented and evaluated to assess its effectiveness in real-time gesture recognition, adaptive learning, and user engagement. The system integrates a mobile-based frontend with a machine learning backend, enabling users to interact with the learning platform through gesture-based inputs and receive immediate feedback.

The sign recognition model demonstrated strong performance in identifying Sinhala sign language gestures. During evaluation, the Convolutional Neural Network (CNN) achieved high classification accuracy across a dataset consisting of 25 Sinhala static letters and 22 dynamic signs. The use of preprocessing techniques such as brightness normalization and contrast enhancement contributed to improved consistency in predictions, even under varying lighting conditions. Additionally, the incorporation of multiple frame analysis was particularly effective for dynamic sign detection, helping to reduce prediction instability and ensuring smoother and more reliable real-time recognition.

The reinforcement learning-based adaptive module showed significant effectiveness in personalizing the learning experience. Users were guided through different levels of difficulty based on their performance, and the system dynamically adjusted lesson complexity. Learners who consistently performed well were introduced to more advanced signs, while those facing difficulties received simplified tasks and additional practice opportunities. This adaptive behavior resulted in a more balanced and user-friendly learning experience.

The spaced repetition mechanism also contributed positively to knowledge retention. Signs that were frequently forgotten were automatically reintroduced at appropriate intervals, reinforcing long-term memory. Users were able to revisit previously learned gestures without feeling overwhelmed, as the system intelligently scheduled review sessions based on individual performance patterns.

From a frontend perspective, the system provided a responsive and interactive user experience. The mobile application successfully delivered real-time feedback through visual indicators, such as success and error messages, along with actionable suggestions for improvement. The integration of camera-based gesture capture enabled seamless interaction, while the use of efficient state management ensured smooth transitions between learning stages.

Performance testing indicated that the system maintained acceptable response times for real-time feedback. The use of optimized image processing and frame-skipping techniques reduced computation overhead, allowing the application to run efficiently on standard mobile devices. Additionally, the system demonstrated stability during continuous usage, with no significant memory leaks or performance degradation observed.

Overall, the results indicate that the system successfully meets its design objectives by providing accurate gesture recognition, adaptive learning capabilities, and engaging user experience.

3.2 Research Findings

The evaluation of the proposed Adaptive Sinhala Sign Language Learning System revealed several significant findings related to adaptive learning effectiveness, machine learning performance, and user engagement within an interactive educational environment. One of the most important findings is that the integration of adaptive learning mechanisms substantially improves user performance compared to traditional static learning approaches. Unlike fixed systems that present the same content to all users, the proposed system continuously monitors user behavior and dynamically adjusts lesson difficulty. This ensures that learners operate within an optimal challenge range, where tasks are neither too easy nor excessively difficult. As a result, users experienced smoother learning progression, with reduced frustration levels and increased motivation. It was observed that learners were more likely to complete lessons and maintain consistency in practice when the system adapted to their individual capabilities. Another key finding highlights the effectiveness of real-time feedback in accelerating the learning process. Immediate feedback allows users to quickly identify mistakes and make necessary corrections without delay. The system goes beyond simple correct/incorrect indicators by providing context-aware feedback, including suggestions related to hand positioning, gesture clarity, and movement accuracy. This type of feedback was found to significantly improve user understanding and skill development. Learners were able to refine their gestures more effectively, leading to higher accuracy rates over time.

The study also demonstrates the strong impact of combining machine learning with reinforcement learning for personalization. The reinforcement learning component enabled the system to learn from user interactions and continuously refine its decision-making strategies. Over time, the system developed more accurate predictions regarding appropriate difficulty levels, feedback types, and review schedules. This resulted in highly personalized learning paths, where each user experienced a unique sequence of lessons

based on their performance. The findings confirm that reinforcement learning can effectively enhance intelligent tutoring systems by enabling continuous adaptation and improvement. Another important observation is the effectiveness of supporting both 25 Sinhala static letters and 22 dynamic signs within a single system. Static signs, which involve fixed hand shapes, were recognized with high accuracy due to their relatively stable visual features. However, the inclusion of dynamic signs, which involve motion and transitions, introduced additional complexity. The system successfully addressed this challenge by analyzing multiple frames and ensuring temporal consistency in predictions. This approach reduced recognition errors and improved reliability when detecting motion-based gestures. The ability to handle both static and dynamic signs significantly enhances the practical applicability of the system. The incorporation of spaced repetition was found to play a crucial role in improving long-term knowledge retention. Instead of relying on one-time learning, the system review sessions based on user performance and forgetting patterns. Signs that were frequently answered incorrectly were reintroduced at optimal intervals, reinforcing memory retention. Users demonstrated improved recall ability over time, indicating that spaced repetition is an effective strategy for sustaining learning outcomes in sign language education.

From a data and model perspective, the findings emphasize the importance of dataset quality and preprocessing techniques. The structured dataset, combined with data augmentation methods such as rotation and brightness variation, allowed the model to generalize effectively across different users and environmental conditions. Preprocessing steps, including normalization and resizing, ensured consistency in model input, which contributed to stable and accurate predictions. This highlights that careful data preparation is essential for achieving reliable machine learning performance. User engagement analysis provided additional insights into system effectiveness. It was observed that learners were more actively involved when the system included visual feedback elements, progress indicators, and performance tracking features. Elements such as accuracy percentages, streak counts, and completion status motivated users to continue practicing. The adaptive nature of the system further contributed to engagement by ensuring that users were consistently challenged without being overwhelmed. Furthermore, system performance and responsiveness were identified as critical factors influencing user satisfaction. Real-time interaction is essential in gesture-based learning systems, and any delay in feedback can negatively affect the user experience. The implementation of performance optimization techniques, such as frame skipping and efficient state management, enabled the system to maintain smooth operation and quick response times. This ensured that users received immediate feedback, which is essential for effective skill acquisition.

Finally, the study indicates that personalization is not a static feature but a continuous process. As users interact with the system over time, more data is collected, allowing the reinforcement learning model to improve its recommendations. This results in progressively better adaptation and a more refined learning experience. The findings suggest that long-term usage of the system leads to increased effectiveness, as the system becomes more aligned with individual learning patterns.

In summary, the research findings confirm that the integration of adaptive learning, real-time feedback, machine learning, and reinforcement learning creates a highly effective and engaging sign language learning environment. The system not only improves accuracy and retention but also enhances user motivation and interaction, making it a strong solution for modern educational applications.

3.3 Discussion

The results and findings of this study clearly demonstrate the effectiveness of integrating machine learning, reinforcement learning, and adaptive learning strategies within a sign language education system. The proposed Adaptive Sinhala Sign Language Learning System successfully addresses several critical limitations found in traditional learning platforms, particularly the lack of personalization, limited feedback mechanisms, and absence of real-time interaction. By combining these advanced technologies, the system provides a more intelligent, responsive, and user-centered learning experience. One of the most significant strengths of the system lies in its ability to adapt to individual learners. Traditional systems typically follow a fixed curriculum, where all users are exposed to the same content regardless of their performance or learning pace. In contrast, the proposed system continuously monitors user behavior and dynamically adjusts lesson difficulty, feedback, and progression. This ensures that learners remain within an optimal learning zone, where they are appropriately challenged without being overwhelmed. The reinforcement learning model plays a key role in this process by learning from user interactions and refining its decision-making over time. This continuous adaptation enhances the overall effectiveness of the learning process, making it particularly suitable for skill-based domains such as sign language, where repeated practice and timely feedback are essential.

Another important aspect highlighted in this study is the integration of real-time gesture recognition. The ability to detect and classify both 25 Sinhala static letters and 22 dynamic signs using camera-based input significantly enhances the interactivity of the system. Unlike traditional learning methods that rely on passive observation, this

approach encourages active participation, allowing users to practice gestures and receive immediate feedback. The inclusion of dynamic sign recognition further strengthens the system, as it reflects more realistic communication scenarios where motion and transitions are involved. This makes the learning experience more practical and aligned with real-world usage. The role of feedback in the system also deserves special attention. The study confirms that providing immediate, context-aware feedback greatly improves learning outcomes. Instead of offering simple correctness indicators, the system delivers meaningful suggestions that guide users toward improvement. This personalized feedback mechanism helps learners understand their mistakes more clearly and adjust their performance accordingly. Additionally, motivational feedback elements contribute to maintaining user engagement, which is a crucial factor in long-term learning success.

Despite these strengths, the study also identifies several limitations that need to be considered. One of the primary challenges is the dependency of the sign recognition model on environmental conditions. Factors such as lighting, background complexity, and camera quality can influence the accuracy of predictions. Although preprocessing techniques such as brightness normalization and contrast enhancement help reduce these effects, variability in real-world conditions can still impact performance. This suggests that further improvements, such as incorporating more diverse training data or using advanced deep learning architectures, could enhance the robustness of the system. Another limitation relates to the computational requirements of real-time processing. The system relies on continuous image capture, preprocessing, and model inference, which can be resource-intensive. While optimization techniques such as frame skipping and image compression were implemented to improve efficiency, performance may still vary depending on the hardware capabilities of the user's device. Ensuring consistent performance across a wide range of devices, especially low-end smartphones, remains an important area for future improvement.

The reinforcement learning component, although highly beneficial, also presents certain challenges. In the early stages of system usage, the model may not have sufficient data to make highly accurate decisions. During this phase, the system relies more on exploration, which can result in less optimal adaptations. However, as more interaction data is collected, the model gradually improves its performance and provides more accurate personalization. This indicates that the system becomes more effective over time, highlighting the importance of long-term usage for achieving maximum benefits.

Another point of discussion is the scalability of the system. While the current implementation focuses on a specific set of Sinhala signs, the overall architecture is flexible and can be extended to support additional signs, words, or even other sign

languages. This scalability makes the system a promising foundation for future developments in sign language education and assistive communication technologies. However, expanding the system would require additional data collection, model training, and validation to maintain accuracy and reliability.

Furthermore, user engagement plays a critical role in the success of the system. The study shows that features such as progress tracking, visual feedback, and adaptive difficulty contribute significantly to maintaining user interest. The interactive nature of the application encourages users to practice more frequently, which directly impacts learning outcomes. This highlights the importance of combining technical innovation with user-centered design principles when developing educational applications. In addition, the inclusion of spaced repetition enhances the long-term effectiveness of the system. By scheduling reviews based on individual performance, the system ensures that learners do not forget previously learned signs. This approach aligns with established learning theories and reinforces the importance of memory-based strategies in skill development. The combination of adaptive learning and spaced repetition creates a comprehensive learning framework that supports both short-term improvement and long-term retention.

In conclusion, the proposed Adaptive Sinhala Sign Language Learning System demonstrates strong potential as an effective and scalable educational tool. The integration of real-time gesture recognition, reinforcement learning, and personalized feedback creates a powerful and engaging learning environment. While certain limitations exist, they do not outweigh the overall benefits of the system. With further improvements in model robustness, performance optimization, and dataset expansion, the system can be enhanced to achieve even greater effectiveness. This research highlights the value of intelligent, adaptive technologies in modern education and provides a strong foundation for future advancements in sign language learning systems.

4. CONCLUSION

This research presented the design, development, and evaluation of an Adaptive Sinhala Sign Language Learning System aimed at improving the effectiveness, accessibility, and engagement of sign language education. The system was developed to address key limitations observed in traditional learning approaches, such as the lack of personalization, limited feedback mechanisms, and insufficient support for long-term knowledge retention. By integrating machine learning, reinforcement learning, and adaptive learning strategies, the proposed system provides a modern, intelligent solution that enhances both learning outcomes and user experience. One of the primary contributions of this research is the successful implementation of a real-time sign recognition system capable of identifying 25 Sinhala static letters and 22 dynamic signs. The use of a Convolutional Neural Network (CNN) enabled the system to accurately classify hand gestures by extracting spatial features from input images. Additionally, the incorporation of temporal analysis for dynamic signs improved prediction stability and ensured reliable recognition of motion-based gestures. The results demonstrate that the system is capable of performing accurate gesture recognition under controlled conditions, making it a strong foundation for practical applications in sign language learning.

Another significant contribution of this study is the integration of reinforcement learning to create a fully adaptive learning environment. Unlike traditional systems that rely on predefined lesson structures, the proposed system continuously learns from user interactions and adjusts its behavior accordingly. By analyzing factors such as accuracy, response time, confidence scores, and learning history, the system dynamically determines the most appropriate learning path for each user. This includes adjusting difficulty levels, selecting suitable feedback, and scheduling review sessions. As a result, each learner experiences a personalized journey that aligns with their individual pace and capabilities. The incorporation of spaced repetition further enhances the effectiveness of the system by supporting long-term knowledge retention. Instead of focusing solely on short-term performance, the system ensures that previously learned signs are revisited at optimal intervals. This approach reduces forgetting and strengthens memory consolidation, which is essential for mastering sign language. The combination of adaptive learning and spaced repetition creates a balanced learning framework that supports both immediate improvement and sustained knowledge retention.

From a user experience perspective, the system provides an interactive and engaging platform for learning. The mobile application interface enables users to practice gestures in real time using the device camera, making the learning process more practical and immersive. The inclusion of visual feedback, progress tracking, and motivational elements

helps maintain user interest and encourages continuous practice. The frontend design, supported by efficient state management and responsive components, ensures smooth interaction and usability across different learning stages.

The feedback mechanism implemented in the system plays a crucial role in guiding learners. By providing immediate and context-aware feedback, the system allows users to identify mistakes and improve their performance effectively. The use of personalized feedback, generated through the adaptive learning engine, ensures that learners receive guidance that is relevant to their current skill level. This targeted feedback approach significantly enhances the learning process compared to traditional systems that offer only basic correctness indicators.

Despite the strengths of the system, several limitations were identified during the research. One of the main challenges is the dependency of the sign recognition model on environmental factors such as lighting conditions, background complexity, and camera quality. Although preprocessing techniques help mitigate these issues, variations in real-world environments can still affect model performance. Future improvements could involve the use of larger and more diverse datasets, as well as advanced deep learning models, to enhance robustness and generalization. Another limitation is related to the computational requirements of real-time processing. The system relies on continuous image capture and analysis, which can be resource-intensive, particularly on low-end devices. While optimization techniques such as frame skipping and image compression were implemented, ensuring consistent performance across all devices remains a challenge. Addressing this issue may require further optimization or the use of lightweight models designed for mobile environments.

The reinforcement learning component also presents certain challenges, particularly during the initial stages of system usage. Since the model relies on interaction data to learn optimal strategies, early recommendations may not always be fully accurate. However, as more data is collected over time, the system becomes more effective in personalizing the learning experience. This highlights the importance of continuous usage for achieving the full benefits of the adaptive learning system. In addition to these limitations, the current system focuses on a specific set of Sinhala signs, which may limit its immediate applicability to broader sign language communication. However, the architecture is designed to be scalable, allowing for the inclusion of additional signs, words, and even other sign languages in future developments. Expanding the dataset and enhancing model capabilities would enable the system to support more comprehensive learning and communication needs. The findings of this research demonstrate that the integration of intelligent technologies in educational systems can significantly improve learning

outcomes. The use of machine learning enables accurate recognition of user inputs, while reinforcement learning introduces adaptability and personalization. Together, these technologies create a powerful learning environment that responds to user needs in real time. This approach represents a shift from traditional, static learning methods to dynamic, user-centered educational systems.

Furthermore, the system has potential applications beyond individual learning. It can be used as a supportive tool in educational institutions, helping teachers deliver more interactive and effective sign language lessons. It may also serve as an assistive technology for individuals with hearing impairments, facilitating communication and improving accessibility. By providing an accessible and scalable platform, the system contributes to the broader goal of inclusive education.

In conclusion, the proposed Adaptive Sinhala Sign Language Learning System successfully achieves its objectives by combining real-time gesture recognition, adaptive learning, and personalized feedback into a single platform. The system addresses key challenges in sign language education and provides an effective solution that enhances both learning efficiency and user engagement. While there are areas for improvement, the overall performance and adaptability of the system demonstrate its strong potential for real-world application.

Future work can focus on expanding the dataset, improving model accuracy under diverse conditions, and optimizing system performance for a wider range of devices. Additionally, integrating advanced technologies such as deep reinforcement learning and multimodal interaction could further enhance the capabilities of the system. With continued development and refinement, the proposed system has the potential to become a comprehensive and widely adopted solution for sign language learning and communication.

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6. GLOSSARY

Adaptive Learning: A learning approach where the system adjusts lesson difficulty, feedback, and progression based on the learner's performance.

Artificial Intelligence (AI): Technology that enables systems to perform tasks such as learning, decision-making, and pattern recognition.

Machine Learning (ML): A subset of AI that allows systems to learn from data and improve performance without explicit programming.

Reinforcement Learning (RL): A machine learning technique where a system learns optimal actions by receiving rewards or penalties based on user performance.

Convolutional Neural Network (CNN): A deep learning model used for image recognition, applied in this system to identify sign language gestures.

Gesture Recognition: The process of detecting and interpreting hand movements using computer vision techniques.

Static Signs: Sign language gestures that involve fixed hand shapes without movement.

Dynamic Signs: Sign language gestures that involve motion and transitions over time.

Dataset: A collection of images used to train and evaluate the gesture recognition model.

Data Augmentation: Techniques used to increase dataset diversity by applying transformations such as rotation and brightness changes.

Preprocessing: The process of preparing input data (e.g., resizing, normalization) before feeding it into the model.

Confidence Score: A value that indicates how certain the model is about its prediction.

Real-Time Processing: The ability of the system to process input and provide feedback instantly.

Spaced Repetition: A learning technique that schedules reviews at specific intervals to improve memory retention.

State Representation: A set of parameters (e.g., accuracy, response time) used to describe the learner's current performance.

Reward Function: A mechanism in reinforcement learning that assigns values to actions based on their outcomes.

Adaptive Feedback: Personalized feedback provided based on the learner's performance and behavior.

Client-Server Architecture: A system structure where the mobile application communicates with a backend server for processing and data handling.

API (Application Programming Interface): A communication interface that allows the frontend and backend systems to interact.

Frontend: The user-facing part of the system where interaction and learning take place.

Backend: The server-side component responsible for processing data and running machine learning models.